

Market Power Becomes Wedges for Misallocation

A Review of the Literature

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Abstract

This review shows that, once market power is allowed to *vary* across plants, these are not two subjects but one. A markup or a markdown is a wedge; a wedge that differs across plants misallocates resources; therefore the dispersion of market power *is* a form of misallocation, and the two literatures must be read together. The argument turns entirely on the demand system. Under the constant-elasticity (CES) benchmark that underlies the canonical Hsieh–Klenow model, every plant charges the same markup, market power is uniform, and it causes *no* misallocation—a knife-edge case. We therefore develop, equation by equation, the demand systems that break this knife-edge and deliver *variable* markups: the Kimball flexible aggregator, translog and other variable-elasticity preferences, the linear demand of Melitz & Ottaviano (2008), and the oligopoly model of Atkeson & Burstein (2008) in which the markup rises with a plant’s market share. For each we state the economic meaning, explain how the markup and its variation are estimated (pass-through and strategic complementarities), and how the estimates are read. We then set out the formal theory that joins market power to aggregate productivity: the level-versus-dispersion result of Peters (2020) (aggregate TFP depends on the *dispersion* of markups, factor prices on their *mean*); the general-equilibrium aggregation of Baqaee & Farhi (2020) (allocative efficiency as the covariance of markups with reallocation, and a ~15% TFP cost of markup dispersion); and

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the welfare decomposition of Edmond, Midrigan, & Xu (2023) into an aggregate-markup channel, a misallocation channel, and an entry channel. We carry the same logic to the factor market, where the *markdown* is the wedge and its dispersion is the misallocation of labour that is the thesis’s central finding. A dedicated section shows why markups and markdowns cannot be measured separately—the production-approach “markup” is the product of the two (Ren & Zhang, 2025)—and why this makes the joint reading unavoidable. Finally we state precisely the condition under which *equalising the markdown* would deliver the efficiency gains Review B quantified: the markdown must be both dispersed and correlated with productivity, exactly the Restuccia & Rogerson (2008) correlated-distortion logic. The review closes on the open questions that Review D, on macroeconomic and trade determinants, takes up.

Keywords: variable markups; wage markdowns; demand systems; misallocation; monopolistic competition; oligopoly; Kimball preferences; pass-through; Indian manufacturing.

JEL: D43, L11, L13, D24, O47, L60.

1 Introduction: two literatures, one phenomenon

The thesis has, to this point, kept two accounts. Review A measured *market power*: the gross markup $\mu_{it} = P_{it}/MC_{it}$ of price over marginal cost in the product market, and the wage markdown $\psi_{it} = MRPL_{it}/w_{it}$ of the marginal revenue product of labour over the wage in the factor market. Review B measured *misallocation*: the dispersion of marginal revenue products across plants and the counterfactual output gain from equalising them. These have historically been separate research programmes—one in industrial organisation and the economics of monopoly and monopsony, the other in macro-development and growth accounting—with separate data, separate models, and separate journals.

This review argues that the separation is an artefact of a single, and singular, modelling assumption: constant-elasticity (CES) demand. Under CES every plant faces the same price-elasticity of demand and therefore charges the *same* markup; market power exists but does not *vary*, and—because only the dispersion of a wedge misallocates (Review B, §2)—uniform market power causes no misallocation at all. The moment demand departs from CES, markups vary across plants; a varying markup is a dispersed revenue wedge; and a dispersed revenue wedge is precisely misallocation. The two literatures collapse into one. The same statement holds on the factor side with the markdown in place of the markup. *The purpose of this review is to develop, carefully and equation by equation, the theory that joins market power to misallocation, and to state the exact conditions under which the thesis’s finding—that the dispersion of the wage markdown is the leading source*

of labour misallocation in Indian manufacturing—holds and is quantitatively what it is measured to be.

The plan is as follows. Section 2 restates the wedge logic of Review B in the language of markups and markdowns, fixing the efficiency benchmark. Section 3 develops the general theory of variable markups—the markup as a function of the demand elasticity, and the “superelasticity” that governs how fast it moves. Section 4 works through the alternative demand systems one by one (CES, Kimball, translog, linear, oligopoly), each with its markup equation, its route to the data, and the reading of its estimate. Section 5 sets out the formal models in which market power *generates* misallocation: Peters (2020), Baqaee & Farhi (2020), Edmond, Midrigan, & Xu (2023), and the factor-market analogue that carries the argument to the wage markdown. Section 6 shows why markups and markdowns are not separately identified and why a joint reading is forced. Section 7 states the condition under which equalising the markdown delivers the Review B efficiency gains. Section 8 reviews the evidence—global, Indian, and the thesis’s own—and Section 9 hands over to Review D.

2 The wedge logic restated: markups and markdowns as the market-power wedges

2.1 Two wedges, one accounting

Review B wrote distortions as *wedges* inserted into the plant’s first-order conditions. Market power is a wedge of a particular, behavioural kind. Cost minimisation with a product-market markup μ_{si} and a factor-market markdown ψ_{si} gives the two conditions that connect this review to the previous two:

$$P_{si} \left(1 - \frac{1}{\varepsilon_{si}^D}\right) = MC_{si}, \quad \mu_{si} \equiv \frac{P_{si}}{MC_{si}} = \frac{\varepsilon_{si}^D}{\varepsilon_{si}^D - 1}, \quad \text{MRPL}_{si} = \psi_{si} w_{si}, \quad \psi_{si} = 1 + \frac{1}{\varepsilon_{si}^L}. \quad (1)$$

Here ε_{si}^D is the price-elasticity of demand the plant perceives in its *output* market and ε_{si}^L is the elasticity of labour supply it faces in its *input* market.

Economic meaning. The two equations are mirror images. In the product market the plant restricts output until marginal revenue equals marginal cost, opening a gap $\mu_{si} > 1$ between price and cost; in the labour market it restricts hiring until the marginal revenue product exceeds the wage, opening a gap $\psi_{si} > 1$ between what the last worker earns the firm and what the firm pays. Both gaps are governed by an *elasticity*: the markup is large when demand is inelastic (ε^D small), the markdown is large when labour supply is inelastic (ε^L small). Market power in *either* market is a wedge, and the whole of this review is about what happens when these elasticities—and hence the wedges—differ across plants.

How it is taken to data. The markup is recovered by the production approach of Review A ($\mu = \theta_M/\alpha_M$, the output elasticity of a flexible input over its revenue share) and the markdown by the Brooks-et-al. ratio ($\psi = \mu_L/\mu_M$). Neither requires the analyst to observe the demand or supply elasticity directly; the elasticities are *inferred* from the wedges, exactly as the Hsieh–Klenow wedges were inferred from marginal revenue products.

How it is interpreted. A plant with a high μ_{si} commands product-market power; a plant with a high ψ_{si} commands labour-market power. The questions of this review are second-moment questions: how dispersed are these wedges, and what does their dispersion do to aggregate productivity?

2.2 The efficiency benchmark: only dispersion misallocates

The efficient allocation equalises marginal revenue products across plants within an industry. Write the plant’s labour wedge as $\tau_{si}^L = \psi_{si} - 1$ and its revenue wedge as $\tau_{si}^R \propto \mu_{si} - 1$. The reallocation gain from moving to the efficient allocation depends only on how these wedges are *distributed*: a wedge common to all plants,

$$\mu_{si} = \bar{\mu} \text{ for all } i, \quad \psi_{si} = \bar{\psi} \text{ for all } i, \quad (2)$$

rescales every plant’s marginal revenue product by the same factor and leaves *relative* allocations—and therefore aggregate TFP—unchanged.

Economic meaning. This is the single most important idea in the review and the reason the two literatures connect only through *variable* market power. A uniform markup is a uniform tax on output: it makes the economy smaller (too little is produced overall) but not *mis*-allocated (what is produced is produced by the right plants). A uniform markdown likewise depresses the wage bill everywhere but does not move labour to the wrong plants. Misallocation—in the strict, Review B sense of inputs in the wrong hands—requires *dispersion* in the wedge.

How it is taken to data. One computes the cross-plant variance $\text{Var}(\ln \mu_{si})$ and $\text{Var}(\ln \psi_{si})$ within industry×year. It is these variances, not the means, that the misallocation calculations of Section 5 require.

How it is interpreted. The distinction cleaves the welfare cost of market power into two economically different pieces—a *level* cost (the economy is too small, borne even under uniform market power) and a *dispersion* cost (the economy is misallocated, borne only under variable market power). Section 5 shows that the leading models of the linkage are, at bottom, formalisations of exactly this split; and the thesis’s contribution lives entirely in the second piece, the dispersion cost, on the factor side.

3 The general theory of variable markups

3.1 The markup as a function of the demand elasticity

Everything begins with the profit-maximising markup already stated in (1):

$$\mu_{si} = \frac{\varepsilon_{si}^D}{\varepsilon_{si}^D - 1}, \quad \varepsilon_{si}^D = -\frac{\partial \ln q_{si}}{\partial \ln p_{si}} > 1. \quad (3)$$

Economic meaning. The markup is a decreasing function of the perceived demand elasticity: the harder it is for customers to substitute away (low ε^D), the more the plant can raise price above cost. *Whether markups vary across plants is therefore entirely a question of whether the demand elasticity ε_{si}^D varies across plants.* If ε^D is a constant σ —the CES case—then every plant charges the identical markup $\sigma/(\sigma - 1)$ and, by (2), market power causes no misallocation. Variable markups, and hence the entire linkage this review develops, exist if and only if the demand system makes ε^D depend on the plant’s own price, size, or market share.

How it is taken to data. Two routes. The *production approach* recovers μ_{si} plant-by-plant without a demand system ($\mu = \theta_M/\alpha_M$) and then *tests* whether the recovered markup covaries with size or market share—if it does, demand is non-CES. The *demand approach* estimates a demand system directly (below) and reads ε_{si}^D , hence μ_{si} , off the estimated elasticity. The two are complementary: the first is agnostic about demand, the second is explicit about it.

How it is interpreted. A markup that rises with plant size or market share is the empirical signature of a pro-competitive demand system (Kimball, translog, oligopoly): big plants face less-elastic residual demand and mark up more. This single fact—documented for many countries and, in the thesis, for India—is what forces the misallocation and market-power questions to be answered together.

3.2 The superelasticity and pass-through: how fast the markup moves

CES is not merely a special value of the elasticity; it is the knife-edge at which the elasticity does not move at all. The object that measures the movement is the *superelasticity*—the elasticity of the demand elasticity itself. Its most useful empirical avatar is the *pass-through* of marginal cost into price:

$$\rho_{si} \equiv \frac{\partial \ln p_{si}}{\partial \ln MC_{si}} = \frac{1}{1 + \Gamma_{si}}, \quad \Gamma_{si} \equiv -\frac{\partial \ln \varepsilon_{si}^D}{\partial \ln q_{si}} \text{ (the superelasticity)}, \quad (4)$$

so that complete pass-through $\rho_{si} = 1$ corresponds to a constant markup ($\Gamma_{si} = 0$, the CES case) and incomplete pass-through $\rho_{si} < 1$ corresponds to markups that *rise* when the plant expands.

Economic meaning. When a plant’s marginal cost falls (say through a productivity gain or a cheaper imported input), a CES plant passes the entire cut to price and holds its markup fixed; a plant with a pro-competitive demand system instead *absorbs part of the cut as a higher markup*, so the price falls by less than cost. Incomplete pass-through is thus the same phenomenon as size-increasing markups, seen dynamically. This is also why more productive plants end up with higher markups—a fact that, as Section 5 shows, makes market power *correlated* with productivity and therefore especially consequential for misallocation.

How it is estimated. Pass-through and its cross-plant heterogeneity are estimated by regressing changes in prices on changes in marginal cost (or in cost shifters such as exchange rates and input tariffs), and on competitors’ prices to capture strategic complementarity. Amiti, Itskhoki, & Konings (2019), using matched producer data, estimate that a typical firm responds to its competitors’ price changes with an elasticity of about 0.4 and to its own cost shocks with an elasticity of about 0.6; crucially, these responses are *heterogeneous*—small firms show near-complete pass-through and no strategic complementarity (they behave CES), while large firms show strong complementarity and marked markup variability. De Loecker et al. (2016) find the same mechanism in Indian trade liberalisation: firms met falling input costs by *raising* markups rather than cutting prices one-for-one.

How it is interpreted. A pass-through below one, concentrated among large plants, is direct evidence that markups are variable and increasing in size, that the CES benchmark fails, and therefore that market-power dispersion is a live source of misallocation. The degree of incompleteness $(1 - \rho)$ is a summary of how far the demand system departs from CES.

4 Alternative demand systems, equation by equation

We now take the five workhorse demand systems in turn. In each case we give the demand (or aggregator) equation, the implied markup, its economic meaning, how the system is estimated, and how its markup estimate is read. The systems are ordered from the knife-edge CES outward.

4.1 Constant elasticity of substitution (CES): the knife-edge

The benchmark, inherited from Review B, aggregates varieties with a constant elasticity σ and delivers isoelastic demand:

$$Y_s = \left(\sum_i Y_{si}^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}, \quad \frac{q_{si}}{Q_s} = \left(\frac{p_{si}}{P_s} \right)^{-\sigma}, \quad \mu_{si} = \frac{\sigma}{\sigma-1} \text{ (constant)}. \quad (5)$$

Economic meaning. Every plant, whatever its size, faces the same constant demand elasticity σ and charges the identical markup $\sigma/(\sigma - 1)$. Market power is present ($\mu > 1$) but perfectly uniform, so by (2) it produces no misallocation: this is why the canonical Hsieh–Klenow model can treat all revenue-productivity dispersion as *distortions* and ignore markups. CES is analytically irresistible but empirically a knife-edge, contradicted by the pass-through evidence of §3.

How it is estimated. σ is imported (typically 3–5), not estimated within the model; given σ , everything else follows. *How it is interpreted:* under CES the entire apparatus of variable markups is switched off, and the market-power and misallocation literatures remain formally separate. The rest of this section restores the connection by relaxing CES.

4.2 The Kimball flexible aggregator

Kimball (1995) generalises CES by defining the output aggregator implicitly through a smooth, increasing, concave function Υ :

$$\sum_i \Upsilon\left(\frac{q_{si}}{Q_s}\right) = 1, \quad \varepsilon_{si}^D = \varepsilon^D\left(\frac{q_{si}}{Q_s}\right) \text{ decreasing in } \frac{q_{si}}{Q_s}, \quad \mu_{si} = \frac{\varepsilon_{si}^D}{\varepsilon_{si}^D - 1} \text{ increasing in size.} \quad (6)$$

Economic meaning. The Kimball aggregator breaks the CES knife-edge in the mildest possible way: it keeps monopolistic competition (a continuum of small plants, each ignoring its effect on the aggregates) but lets the demand elasticity decline as a plant grows relative to the market. Larger plants therefore face less-elastic demand and charge higher markups; the speed at which the markup rises with size is exactly the superelasticity Γ of (4). CES is the special case $\Upsilon(x) = x^{(\sigma-1)/\sigma}$, for which $\Gamma = 0$.

How it is estimated. The Kimball system is parameterised by two numbers—an elasticity level and a superelasticity—which are pinned down by matching (i) an average markup and (ii) the degree of incomplete pass-through or the slope of markups in size (Klenow & Willis, 2016). In practice one estimates pass-through or the markup–size gradient (as in §3) and inverts for the superelasticity.

How it is interpreted. A larger estimated superelasticity means markups fan out more steeply with size, so a given size distribution generates more markup *dispersion* and hence more misallocation. The Kimball model is the canonical vehicle for “real rigidities” in macro and, for this thesis, the cleanest monopolistic-competition device for turning size heterogeneity into markup dispersion.

4.3 Translog and variable-elasticity-of-substitution preferences

A second route to variable markups works through expenditure shares rather than quantities. With symmetric translog preferences the demand elasticity depends on a

plant’s revenue share s_{si} :

$$\ln s_{si} = \beta_i + \sum_j \gamma_{ij} \ln p_{sj}, \quad \varepsilon_{si}^D = 1 - s_{si} + \frac{b}{s_{si}}, \quad b \equiv -\gamma_{ii} > 0, \quad \mu_{si} = \frac{\varepsilon_{si}^D}{\varepsilon_{si}^D - 1} \text{ increasing in } s_{si}. \quad (7)$$

Economic meaning. As a plant’s revenue share s_{si} rises, the term b/s_{si} shrinks and the elasticity ε_{si}^D falls towards a floor, so the markup rises: a dominant plant faces inelastic residual demand and marks up heavily, while an atomistic plant ($s_{si} \rightarrow 0$) faces near-infinite elasticity and prices near marginal cost. Translog belongs to the broader class of variable-elasticity-of-substitution (VES) preferences of Zhelobodko et al. (2012), in which whether markups rise or fall with size is governed by the curvature of the sub-utility function.

How it is estimated. The share equations in (7) are estimated directly as a demand system (the parameters γ_{ij} from regressions of expenditure shares on log prices), which is how Feenstra & Weinstein (2017) recover endogenous markups and quantify the welfare effect of globalisation through changing markups and variety.

How it is interpreted. Because the markup is tied to the observable revenue share, translog makes the market-power-misallocation link operational with standard data: the cross-plant dispersion of shares maps directly into a dispersion of markups, and hence—by Section 5—into a measurable misallocation. Whether reallocation raises or lowers welfare then depends, as Dhingra & Morrow (2019) show for general variable-markup preferences, on how markups covary with productivity: the market equilibrium is efficient only in the CES case and is generically distorted otherwise.

4.4 Linear demand: the Melitz–Ottaviano model

Melitz & Ottaviano (2008) generate variable markups from a quasi-linear quadratic utility that yields *linear* demand, so that the markup is an *absolute* margin that depends on the toughness of local competition:

$$p_{si} = c_{si} + (\text{markup})_{si}, \quad \mu_{si}^{\text{abs}} \equiv p_{si} - c_{si} = \frac{1}{2}(p_s^{\text{max}} - c_{si}), \quad (8)$$

where p_s^{max} is the choke price (the price at which demand hits zero) and c_{si} is the plant’s marginal cost.

Economic meaning. Lower-cost (more productive) plants set a larger absolute margin and capture a larger market share, but the *relative* markup p_{si}/c_{si} still varies across plants, and—the hallmark of the model—a larger or more open market lowers the choke price p_s^{max} and compresses every plant’s markup. This is the “pro-competitive effect” of market size and trade: tougher competition squeezes margins. Markups are dispersed (they differ with cost) and endogenous to the competitive environment.

How it is estimated. The choke price and demand slope are estimated from the relation between prices, quantities, and market size across markets or over time; the cost draws c_{si} are recovered from a productivity distribution. *How it is interpreted:* the Melitz–Ottaviano markup gives a transparent account of why market power is not a fixed plant attribute but responds to entry, market size, and trade openness—a bridge directly to the trade determinants of Review D, and a warning that measured markup dispersion partly reflects the competitive environment, not just plant fundamentals.

4.5 Oligopoly: Atkeson–Burstein and markups in market share

The final departure from CES drops monopolistic competition for genuine *oligopoly*, in which large plants internalise their effect on the sector price index. Atkeson & Burstein (2008) nest a within-sector elasticity ρ inside a lower across-sector elasticity η (with $\rho > \eta > 1$) and, under Cournot competition, obtain a perceived elasticity—and hence a markup—that depends on the plant’s within-sector market share s_{si} :

$$\varepsilon_{si}^D = \left[\frac{1 - s_{si}}{\rho} + \frac{s_{si}}{\eta} \right]^{-1}, \quad \mu_{si} = \frac{\varepsilon_{si}^D}{\varepsilon_{si}^D - 1} \text{ increasing in } s_{si}. \quad (9)$$

Economic meaning. A negligible plant ($s_{si} \rightarrow 0$) perceives the high within-sector elasticity ρ and charges the small monopolistic-competition markup $\rho/(\rho - 1)$; a dominant plant ($s_{si} \rightarrow 1$) internalises that raising its price pushes up the whole sector index, perceives only the low across-sector elasticity η , and charges the large markup $\eta/(\eta - 1)$. Markups therefore rise monotonically with market share, and—because shares are dispersed—so are markups. Oligopoly makes market power an increasing function of concentration, tying the markup distribution to the HHI-type measures the thesis already computes in Objective 1.

How it is estimated. Given the two elasticities (ρ, η) and observed market shares, (9) delivers a markup for every plant; the elasticities are calibrated to match trade or pass-through moments (larger firms show lower pass-through, exactly as Amiti, Itskhoki, & Konings (2019) document). *How it is interpreted:* the model predicts that concentrated industries have both higher *and* more dispersed markups, so the dispersion cost of market power should be largest where concentration is highest—a testable cross-industry prediction, and the natural oligopoly counterpart to the oligopsony models that govern the thesis’s labour market (§5).

4.6 Taking stock of the demand systems

The five systems share one message: *CES is the only one in which markups are uniform, and it is the only one in which market power causes no misallocation.* Every empirically serious alternative—Kimball, translog/VES, linear, oligopoly—makes the markup rise with

size or share, so that the observed heterogeneity of plants translates into a heterogeneity of markups, and market-power dispersion becomes a component of misallocation. Which system one adopts changes the *quantitative* mapping from size to markup, but not the *qualitative* conclusion that the two literatures are joined. We now formalise the join.

5 How market power generates misallocation: the formal link

This section states, model by model, the theory that turns a dispersion of markups (or markdowns) into a loss of aggregate productivity. Each result is a formalisation of the level-versus-dispersion split of §2.

5.1 Markups as revenue wedges: the sufficient-statistic parallel

The cleanest bridge to Review B treats the markup as a plant-specific revenue wedge. Because marginal revenue is $P_{si}(1 - 1/\varepsilon_{si}^D) = P_{si}/\mu_{si}$, a plant with a high markup behaves as if it faced an output tax of $1 - 1/\mu_{si}$: it is held below its efficient scale. Feeding this into the Hsieh–Klenow machinery, revenue productivity inherits the markup, and the misallocation attributable to markups is, to the same lognormal approximation as Review B,

$$\ln\left(\frac{Y_s^{\text{efficient}}}{Y_s}\right)\Big|_{\text{markups}} \approx \frac{\sigma}{2} \text{Var}(\ln \mu_{si}). \quad (10)$$

Economic meaning. The output loss from variable markups is, to a first approximation, half the substitution elasticity times the *variance of log markups*—the exact analogue of the TFPR-dispersion result, with the markup playing the role of the wedge. This makes precise the claim of Section 2: it is $\text{Var}(\ln \mu)$, not the level of μ , that misallocates, and the loss vanishes when markups are uniform.

How it is estimated. Recover μ_{si} (production or demand approach), take logs, compute the within-industry variance, and scale by $\sigma/2$. The estimate is reported alongside $\text{Var}(\ln \mu)$ itself, following the same discipline as Review B. *How it is interpreted:* a hypothetical upper bound on the TFP recoverable by equalising markups; it is informative about the *dispersion* cost of market power but says nothing about the *level* cost, which (10) nets out.

5.2 Level versus dispersion in general equilibrium: Peters (2020)

Peters (2020) makes the level/dispersion split exact in a growth model with heterogeneous markups arising from imperfectly competitive output markets. His central aggregation result is that

$$\text{aggregate TFP depends on } \text{Var}(\ln \mu_{si}) \text{ (dispersion);} \quad \text{factor prices depend on } \bar{\mu} \text{ (level).} \quad (11)$$

Economic meaning. This is the general-equilibrium statement of the theme of this review. The *mean* markup determines the labour share and the level of factor prices—the aggregate “output tax” of the level channel—while the *dispersion* of markups determines how far TFP falls short of its efficient level—the misallocation channel. Peters adds a dynamic layer: because markups are set through entry and creative destruction, the misallocation is *endogenous*—more entry intensifies competition, lowering both the level and the dispersion of markups, so that static misallocation and the growth rate are jointly determined. In his Indonesian calibration the dynamic welfare consequences of removing entry barriers are more than four times their static counterpart.

How it is estimated. The markup distribution is recovered from firm-level data; the level and dispersion are computed and mapped, through the model’s closed form, into a factor-price effect and a TFP effect respectively. *How it is interpreted:* the decomposition tells the analyst which margin a policy operates on—whether it changes the average markup (and so factor shares) or its dispersion (and so allocative efficiency)—and warns that the two need not move together.

5.3 The GE cost of markup dispersion: Baqaee–Farhi (2020)

Baqaee & Farhi (2020) provide the nonparametric general-equilibrium aggregation for an economy with arbitrary distortions, markups among them. Their key object is *allocative efficiency*, whose change is a covariance between markups and the reallocation of inputs:

$$d \ln Y = \sum_i \lambda_i d \ln A_i + d \ln \Lambda, \quad d \ln \Lambda > 0 \text{ when inputs flow to higher-markup plants,} \quad (12)$$

where λ_i are the (sales-based) Domar weights and $d \ln \Lambda$ is the change in allocative efficiency.

Economic meaning. Aggregate output growth is technology growth (the first term) plus a reallocation term (the second). Because high-markup plants are held inefficiently small, moving inputs *towards* them raises output—so allocative efficiency improves when market share reallocates to high-markup firms, and the static loss from markups is the distance between the actual allocation and the one that would equate marginal products. On U.S. data Baqaee & Farhi (2020) find that such reallocation towards high-markup firms accounts for about *half* of aggregate TFP growth over 1997–2015, and that eliminating the misallocation caused by the large, dispersed markups in the data would raise aggregate TFP by about *15%*—two orders of magnitude larger than the classic 0.1% welfare-triangle estimate of Harberger (1954), precisely because the cost is one of *dispersion* (misallocation) rather than the *level* (the triangle).

How it is estimated. One measures firm-level markups and Domar weights and applies the nonparametric formulas; the 15% figure is the model-implied TFP gain from removing

markup dispersion. *How it is interpreted:* the number is large but, as the authors stress, sensitive to the imputation of markups—so it is best read as establishing an order of magnitude (markup misallocation is macroeconomically first-order, not a Harberger triangle) rather than a precise point.

5.4 How costly are markups? The three channels of Edmond–Midrigan–Xu

Edmond, Midrigan, & Xu (2023) decompose the total welfare cost of markups, in a dynamic heterogeneous-firm model, into three economically distinct channels:

$$\text{Cost of markups} = \text{aggregate-markup (level)} + \text{misallocation } \left(\text{Var}(\ln \mu_{si}) \right) + \text{inefficient entry.} \quad (13)$$

Economic meaning. The first channel is the level cost of Section 2: even a uniform markup shrinks the economy like an output tax. The second is the dispersion cost: heterogeneous markups misallocate factors across plants. The third is a variety/entry margin absent from the static Hsieh–Klenow world. Calibrated to U.S. manufacturing, they find the *aggregate-markup and misallocation channels dominate* and the entry channel is minor—and that the total cost can be large, with much of it removable by policies (or competition) that compress markup dispersion.

How it is estimated. The markup distribution is calibrated to census moments; each channel is then a model-based counterfactual (equalise markups; set the average markup to one; free entry). *How it is interpreted:* the decomposition is the template the thesis follows on the factor side—separating the level cost of the markdown (a depressed wage bill everywhere) from its dispersion cost (labour in the wrong plants), and concentrating on the latter.

5.5 The factor-market analogue: markdown dispersion and labour misallocation

Everything above has a mirror image in the labour market, and it is the mirror image that the thesis estimates. Replace the product markup μ_{si} by the wage markdown ψ_{si} and the output wedge by the labour wedge. The marginal revenue product of labour then inherits the markdown, and—by the decomposition of Review B—its dispersion is

$$\text{Var}(\ln \text{MRPL}_{si}) = \text{Var}(\ln \psi_{si}) + \text{Var}(\ln w_{si}) + \text{Var}(u_{si}) + 2 (\text{covariances}), \quad (14)$$

so that labour misallocation attributable to labour-market power is governed by $\text{Var}(\ln \psi_{si})$ and its covariance with the rest.

Economic meaning. Just as dispersed markups misallocate output across plants, dispersed markdowns misallocate *labour*: a plant with high wage-setting power hires too few workers

relative to its productivity, and the workers it does not hire go to plants that value them less at the margin. Berger, Herkenhoff, & Mongey (2022) formalise this in a general-equilibrium oligopsony model and find the welfare losses from labour-market power to be large—the factor-market counterpart to the Baqaee & Farhi (2020) product-market losses. The dispersion of the markdown is, literally, the misallocation of labour.

How it is estimated. Compute ψ_{si} (Brooks-et-al. ratio), w_{si} , and MRPL_{si} plant-by-plant; form the variance decomposition (14); the markdown’s contribution is the covariance share $\text{Cov}(\ln \text{MRPL}, \ln \psi) / \text{Var}(\ln \text{MRPL})$. *How it is interpreted:* a large markdown share means labour misallocation *is* the dispersion of labour-market power—the thesis’s headline result, whose magnitude ($\sim 79\%$) and robustness Section 8 records.

6 Why markups and markdowns cannot be measured separately

A practical obstacle now becomes a conceptual point. The production approach does not, by itself, separate product-market power from factor-market power, and understanding why is essential to reading the thesis’s numbers.

6.1 The identity that entangles them

The “markup” recovered from the ratio of an input’s output elasticity to its revenue share is, when the factor market is imperfectly competitive, the *product* of the true product markup and the labour markdown:

$$\ln \mu_{si}^{\text{measured}} = \ln \mu_{si} + \ln \psi_{si}. \quad (15)$$

Economic meaning. A high measured “markup” can reflect product-market power (μ), labour-market power (ψ), or both. The production function alone cannot tell them apart, because both drive a wedge between revenue and the cost of a marginal input. Ren & Zhang (2025) make exactly this decomposition central, showing that a substantial part of the measured rise in U.S. “markups” is in fact a rise in *markdowns*—wage-setting power, not price-setting power. The same identity governs the thesis: what a naive TFPR or markup calculation attributes to the product market may belong to the labour market.

How it is separated. Identification requires a second equation. The Brooks-et-al. strategy uses a *flexible, competitively-priced input* (materials) as a numeraire: since materials are bought competitively, their wedge is the pure product markup, and dividing the labour wedge by the materials wedge isolates the markdown, $\psi = \mu_L / \mu_M$. *How it is interpreted:* the materials-based markup is the product-market wedge; the labour-to-materials ratio is the factor-market wedge; only with both can the analyst say how much of a plant’s total wedge is monopoly and how much is monopsony.

6.2 Variable markups change the measured misallocation

Once markups are variable, they also contaminate the *misallocation* measure of Review B, and correcting for them changes both the level and the counterfactual. Li & Wang (2021) show that treating variable markups as if they were distortions inflates measured misallocation, and that netting them out makes China’s measured misallocation considerably smaller. For India, Liang (2023) shows something subtler and directly relevant: variable markups account for only a *small* share of raw TFPR dispersion once other wedges are allowed, *but* they change the *gain* from a reallocation policy—because more productive firms endogenously choose higher markups, equalising marginal products yields a *smaller* TFP gain than a constant-markup model would predict, so the indirect costs of markups are larger than they first appear.

Economic meaning. Variable markups are not merely a nuisance that inflates dispersion; they alter the returns to reallocation itself, because the plants a planner would expand are the very plants whose markups would then fall. *How it is interpreted for the thesis:* this is why the thesis reports an *integrated gross-output* misallocation measure that nets out the product-market markup using materials as a competitive numeraire (§8), isolating the factor-market wedge whose dispersion it wishes to hold responsible.

6.3 Epifani–Gancia: the whole distribution matters

Epifani & Gancia (2011) generalise the point to a large class of imperfect-competition models and show that it is the *entire distribution* of markups—its second moment in particular—that governs misallocation and welfare, so that any force (trade, policy, demand structure) that widens the markup distribution worsens allocation even if it leaves the average markup unchanged.

How it is interpreted. This validates the thesis’s insistence on dispersion measures over means throughout: the object of interest is $\text{Var}(\ln \mu)$ and $\text{Var}(\ln \psi)$, and a determinant of misallocation is anything that moves these variances—the subject Review D takes up.

7 The condition under which equalising the markdown delivers the gains

We can now state precisely when the thesis’s central counterfactual—“equalise the markdown”—would deliver the efficiency gains Review B quantified. Two conditions must hold, one about dispersion and one about correlation.

7.1 The dispersion (accounting) condition

Start from the Review B decomposition $\ln \text{MRPL}_{si} = \ln \psi_{si} + \ln w_{si} + u_{si}$. Equalising the markdown (setting $\psi_{si} = \bar{\psi}$ for all plants) removes from the variance of $\ln \text{MRPL}$ exactly

the markdown’s covariance share:

$$\frac{\Delta \text{Var}(\ln \text{MRPL})}{\text{Var}(\ln \text{MRPL})} = \frac{\text{Var}(\ln \psi_{si}) + 2 \text{Cov}(\ln \psi_{si}, \ln w_{si} + u_{si})}{\text{Var}(\ln \text{MRPL}_{si})} = \frac{\text{Cov}(\ln \text{MRPL}_{si}, \ln \psi_{si})}{\text{Var}(\ln \text{MRPL}_{si})}. \quad (16)$$

Economic meaning. The fraction of labour misallocation removed by equalising the markdown is the regression (covariance) share of the markdown in the marginal-revenue-product dispersion. If the markdown were the *only* wedge, this share would be one and equalising it would eliminate labour misallocation entirely; if the markdown were orthogonal to MRPL, the share would be zero. Reality lies between, and the share is an object to be *measured*.

How it is estimated. Compute the covariance share directly, or equivalently regress $\ln \text{MRPL}$ on $\ln \psi$ and read the explained share. The thesis finds this share to be about 0.79, with the variance of $\ln \text{MRPL}$ falling by about 0.56 (its standard deviation from 1.20 to 0.80) when the markdown is equalised (§8). *How it is interpreted:* equalising labour-market power would remove roughly four-fifths of the covariance and more than half the variance of the marginal revenue product of labour—most, though not all, of the measured labour misallocation.

7.2 The correlation (Restuccia–Rogerson) condition

Dispersion alone does not determine the *output* cost; the sign of the correlation between the wedge and productivity does. A markdown that is *higher for more productive plants* is far more damaging than a randomly scattered markdown of the same variance, because it actively holds back the plants that should be largest. Formally, the TFP loss rises with

$$\text{Cov}(\ln \psi_{si}, \ln A_{si}) > 0 \implies \text{markdown-driven misallocation is amplified}, \quad (17)$$

which is the labour-market instance of the Restuccia & Rogerson (2008) correlated-distortion result and the Liang (2023) mechanism by which productive firms choose higher markups.

Economic meaning. Equalising the markdown delivers the large Review B gains only if the markdown is both dispersed *and* positively correlated with productivity, so that the plants currently under-hiring are the productive ones that would expand under a competitive labour market. If instead high markdowns fell on low-productivity plants, equalisation would move less output and the gain would be smaller.

How it is estimated. One computes the cross-plant correlation between the markdown (or its dispersion) and a productivity index. The thesis reports a cross-industry correlation of about 0.83 between labour-misallocation (SD $\ln \text{MRPL}$) and markdown dispersion (SD $\ln \psi$)—far above the corresponding correlation with markup dispersion—indicating that where the markdown is most dispersed, labour is most misallocated. *How it is*

interpreted: the joint satisfaction of the dispersion condition (16) and the correlation condition (17) is what licenses the thesis’s claim that equalising the wage markdown would recover the bulk of the efficiency loss from labour misallocation in Indian manufacturing—the precise sense in which market power *is* the misallocation.

8 The empirical record: world, India, and the thesis

8.1 The global picture

The macro evidence now points squarely at variable market power as a first-order source of misallocation. De Loecker, Eeckhout, & Unger (2020) document a rise in the average U.S. markup from 1.21 in 1980 to 1.61 by 2016, driven by a fattening upper tail and a reallocation of activity to high-markup firms—i.e. a widening markup *distribution*, the second moment that Epifani & Gancia (2011) identify as the misallocation-relevant object. Baqaee & Farhi (2020) translate such dispersion into an $\sim 15\%$ TFP cost in general equilibrium; Edmond, Midrigan, & Xu (2023) attribute most of the (large) welfare cost of markups to the aggregate-markup and misallocation channels; and Peters (2020) shows the loss is endogenous to entry and growth. On the estimation side, Amiti, Itskhoki, & Konings (2019) confirm the mechanism at the micro level—incomplete, heterogeneous pass-through consistent with markups rising in size—while cautionary work reminds us that measured markups are sensitive to production-function and demand assumptions (Review A). The synthesis is that market-power dispersion is real, has widened, and is a leading candidate for the misallocation that Review B measures.

8.2 India

For India the linkage is doubly relevant because both product- and labour-market power are pronounced. De Loecker et al. (2016) show, using the 1990s trade liberalisation, that Indian firms exercise variable markups and absorbed input-cost reductions as higher markups rather than lower prices—direct evidence of incomplete pass-through and non-CES demand. Liang (2023) shows for Indian manufacturing that variable markups, while a small share of raw TFPR dispersion, materially reduce the TFP gain from equalising marginal products, because productive firms endogenously choose higher markups. Against this product-market backdrop the thesis’s contribution is to establish that the *labour* market, not the product market, is where the dispersion of market power does most of the misallocative work.

8.3 The thesis’s own linkage results

Applying the joint framework to ASI 2012–13 through 2023–24, the thesis finds that the misallocation of labour in Indian organised manufacturing is, first and foremost, the

dispersion of labour-market power. The variance decomposition (14) assigns about 79% of the variance of $\ln \text{MRPL}$ to the markdown, about 30% to wage dispersion, and about 65% to the net market-power channel (markdown plus product markup). Equalising the markdown alone—the counterfactual of (16)—would cut the variance of $\ln \text{MRPL}$ by about 56% (its standard deviation falling from 1.20 to 0.80). The decomposition is stable over time and across sample cuts, with the markdown share staying in a narrow 0.75–0.84 band, and the cross-industry correlation between labour misallocation and markdown dispersion—the correlation condition (17)—is about 0.83, far above the analogous correlation with markup dispersion. In the language of this review: on Indian data the dispersion condition and the correlation condition are both satisfied for the *markdown* and only weakly for the *markup*, so it is factor-market power, not product-market power, whose dispersion constitutes the misallocation of labour.

Two features make this a genuine test of the linkage rather than a restatement of it. First, the thesis’s *integrated gross-output* misallocation measure nets out the product markup using materials as a competitive numeraire (per (15)), so the labour result is not an artefact of conflating μ with ψ . Second, the markdown share survives the robustness battery (winsorisation, sub-periods, alternative rental rates, alternative cell definitions), which is what one expects of a *structural* wedge and not of a residual. The linkage developed in this review is therefore not a theoretical possibility but the organising fact of the thesis’s data.

9 Open questions and the bridge to Review D

Three questions remain open and set up the final review.

First, which demand system does India actually obey? The thesis’s results are reported under disciplined assumptions, but the exact mapping from size to markup (Kimball vs translog vs oligopoly) is not separately identified in the ASI, and different systems imply different dispersion costs. Estimating pass-through and the markup–share gradient directly on Indian data is a natural next step and would sharpen the quantitative linkage.

Second, why does labour-market power dominate product-market power in the Indian misallocation? The evidence says it does; the *cause*—thin local labour markets, migration frictions, employer concentration, labour regulation—is the province of the determinants analysis. That the markdown is both dispersed and correlated with productivity is a finding in search of a mechanism.

Third, what moves these wedges over time? The markup and markdown distributions are not primitives; they respond to entry, trade, exchange rates, and macroeconomic conditions. Melitz & Ottaviano (2008) and De Loecker et al. (2016) show that trade and market size compress markups; the symmetric question for markdowns—how openness, demand conditions, and factor-market institutions move the *dispersion* of labour-market

power—is exactly the subject of Review D. Review A measured market power; Review B measured misallocation; Review C has shown that, once market power is variable, the dispersion of market power *is* misallocation, and that on Indian data the operative wedge is the wage markdown. Review D turns to the macroeconomic and trade *determinants* that make these wedges rise and fall—the final link from the thesis’s measurements to the forces that drive them.

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